13

Projection and Least Squares

Historical Note

Legendre published the method of least squares in 1805, in an appendix on determining the orbits of comets, and gave it the name *méthode des moindres carrés*. Gauss published his own account in 1809, added the probabilistic justification that makes the method more than a convenient recipe, and remarked that he had been using it since 1795. Legendre was furious, and wrote to say so. The dispute was never settled and probably cannot be: Gauss's notebooks support his claim, and Legendre's grievance about publication is equally sound. What is not in dispute is the calculation that made the method famous. When the asteroid Ceres was lost behind the sun in 1801 after forty-one days of observation, Gauss fitted an orbit to the scattered and mutually inconsistent measurements and predicted where it would reappear; it was found within half a degree of his prediction. Chapter 3 mentioned that episode as the origin of elimination. This chapter says what was actually being solved.

Every system in Chapter 3 was either consistent or hopeless. Real data is almost always the second: more equations than unknowns, measurements that disagree, and no x whatever with $Ax = b$. Refusing to answer is not useful. This chapter asks for the x that comes closest, shows that the question has a clean answer, and shows that the answer is a projection.

13.1 Orthogonal complements

Definition 13.1 (Orthogonal complement)**.** *For a subspace W of an inner product space V,*

$$W^{\perp} = \{v \in V : \langle v, w\rangle = 0 \text{ for every } w \in W\},$$

a subspace of V by Exercise 12.23.

Theorem 13.2 (Every vector splits, in exactly one way)**.** *Let W be a subspace of a finite-dimensional inner product space V. Then every $v \in V$ can be written as*

$$v = w + w', \qquad w \in W, \quad w' \in W^{\perp},$$

and in only one way. Consequently $\dim W + \dim W^{\perp} = \dim V$ *and* $(W^{\perp})^{\perp} = W$.

Proof. By Theorem 12.10, W has an orthonormal basis $e_1, \dots, e_k$. Given v, put

$$w = \sum_{i=1}^{k} \langle v, e_i \rangle \, e_i \in W, \qquad w' = v - w.$$

For each j, $\langle w', e_j \rangle = \langle v, e_j \rangle - \langle v, e_j \rangle = 0$, so w' is orthogonal to every basis vector of W and hence, by linearity, to all of W. That is existence.

For uniqueness, suppose $w_1 + w_1' = w_2 + w_2'$ with $w_i \in W$ and $w_i' \in W^{\perp}$. Then $w_1 - w_2 = w_2' - w_1'$ lies in both W and $W^{\perp}$, so it is orthogonal to itself, and positive definiteness forces it to be $\mathbf{0}$.

For the dimension count, a basis of W together with a basis of $W^{\perp}$ spans V by existence and is independent because the two subspaces meet only in $\mathbf{0}$. Finally $W \subseteq (W^{\perp})^{\perp}$ is immediate from the definition, and the two have the same dimension by applying the count twice, so they are equal by Theorem 6.14(g). □

13.2 Projection onto a subspace

Definition 13.3 (Orthogonal projection)**.** *With notation as above, the map $P_W : V \to V$ sending v to its W-part w is the* orthogonal projection onto W. *Equivalently, $P_W v$ is the unique vector of W with $v - P_W v \in W^{\perp}$, and in an orthonormal basis of W,*

$$P_W v = \sum_{i=1}^{k} \langle v, e_i \rangle \, e_i.$$

The characterisation by $v - P_W v \in W^{\perp}$ mentions no basis, so P_W does not depend on which orthonormal basis was used to compute it. For $\dim W = 1$ this is Definition 2.13 from Chapter 2, written with a unit vector.

Theorem 13.4 (Best approximation)**.** *For every $q \in W$ with $q \neq P_W v$,*

$$\|v - P_W v\| \; < \; \|v - q\| .$$

So $P_W v$ is the point of W closest to v, and it is the only one.

Proof. Write $v - q = (v - P_W v) + (P_W v - q)$. The first bracket lies in $W^\perp$ and the second in W, so they are orthogonal and Pythagoras applies:

$$\|v - q\|^2 = \|v - P_W v\|^2 + \|P_W v - q\|^2 .$$

The last term is strictly positive unless $q = P_W v$. □

Figure 13.1 is the argument as a picture: the residual leaves the plane at a right angle, so any other point of the plane sits at the far end of a hypotenuse.

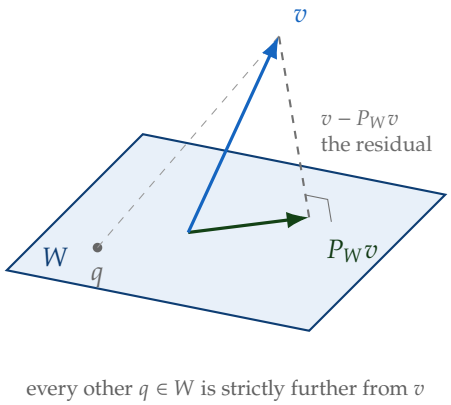


Figure 13.1. Theorem 13.4. The residual $v - P_W v$ leaves the plane at right angles, and that is exactly what makes $P_W v$ the nearest point: for any other q in the plane, the triangle with vertices v, $P_W v$ and q has a right angle at $P_W v$, so the side vq is a hypotenuse and is longer.

Lemma 13.5 ($A^t A$ is invertible when the columns are independent)**.** *Let A be $m \times n$ with linearly independent columns. Then $A^t A$ is invertible.*

Proof. Suppose $A^t A x = \mathbf{0}$. Then $0 = x^t A^t A x = (Ax)^t (Ax) = \|Ax\|^2$, so $Ax = \mathbf{0}$, and independence of the columns gives $x = \mathbf{0}$ by Proposition 6.4. So $N(A^t A) = \{\mathbf{0}\}$ and Theorem 4.16 applies. □

Proposition 13.6 (Projection matrices)**.** *Let $W \subseteq \mathbb{R}^m$ be a subspace.*

(a) If the columns of Q are an orthonormal basis of W, then $P_W = QQ^t$.

(b) *If the columns of A are any basis of W, then $P_W = A(A^tA)^{-1}A^t$.*
(c) *In both cases $P_W^2 = P_W$ and $P_W^t = P_W$.*

Proof. (a) Writing e_i for the columns of Q, we have $QQ^tv = \sum_i e_i\,(e_i^t v) = \sum_i \langle v, e_i\rangle\, e_i$, which is Definition 13.3.

(b) Put $P = A(A^tA)^{-1}A^t$, well defined by Lemma 13.5. For any v, Pv is a combination of the columns of A, hence lies in W; and

$$A^t(v - Pv) = A^tv - (A^tA)(A^tA)^{-1}A^tv = \mathbf{0},$$

so $v - Pv$ is orthogonal to every column of A and therefore to W. By the uniqueness in Definition 13.3, $P = P_W$.

(c) $P^2 = P$ because P_W fixes W pointwise and $Pv \in W$; symmetry is a direct check in either formula, using $(A^tA)^{-1}$ being symmetric. □

Caution

Formula (b) is for understanding, not for computing. It inverts A^tA, and Section 13.4 explains why forming that matrix is the one step to avoid. Note also that P_W is a projection onto W along the perpendicular direction only; matrices with $P^2 = P$ but $P^t \neq P$ project too, but obliquely, and they do not solve the nearest point problem.

13.3 Least squares

Definition 13.7 (Least squares solution)**.** *Given A of size $m \times n$ and $b \in \mathbb{R}^m$, a* least squares solution *of $Ax = b$ is any $\hat{x}$ minimising $\|Ax - b\|$ over all $x \in \mathbb{R}^n$.*

Lemma 13.8. *$C(A)^\perp = N(A^t)$ and $R(A)^\perp = N(A)$.*

Proof. A vector y is orthogonal to every column of A exactly when every entry of A^ty vanishes, which is the first identity. Applying it to A^t and using $C(A^t) = R(A)$ gives the second. □

Theorem 13.9 (The normal equations)**.** *$\hat{x}$ is a least squares solution of $Ax = b$ if and only if*

$$A^tA\,\hat{x} = A^tb.$$

Such an $\hat{x}$ always exists. It is unique exactly when the columns of A are independent, and then $\hat{x} = (A^tA)^{-1}A^tb$.

Proof. As x ranges over $\mathbb{R}^n$, the vector Ax ranges over exactly $C(A)$. So minimising $\|Ax - b\|$ means finding the point of $C(A)$ nearest to b, which by Theorem 13.4 is $P_{C(A)}b$ and is unique. Thus $\hat{x}$ is a least squares solution precisely when $A\hat{x} = P_{C(A)}b$, that is, precisely when $b - A\hat{x} \in C(A)^{\perp}$. By Lemma 13.8 that says $A^t(b - A\hat{x}) = \mathbf{0}$, which is the normal equations.

Existence: $P_{C(A)}b$ lies in $C(A)$, so some x has $Ax = P_{C(A)}b$. Uniqueness: two solutions differ by an element of $N(A)$, which is trivial exactly when the columns are independent, and in that case Lemma 13.5 lets us invert. □

Key Idea

The normal equations say that the residual is orthogonal to every column of A. In the language of data fitting, that is the statement that whatever is left over after the fit contains nothing further that the predictors could have explained. If it did, the fit could be improved by moving along that predictor, so it would not have been the best one. Geometry and statistics are saying the same sentence.

Example 13.10 (Fitting a line to data)**.** *Fit $y = a + bx$ to the four points $(1,1)$, $(2,3)$, $(3,4)$, $(4,4)$. Demanding exactness gives four equations in two unknowns,*

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}, \qquad x = \begin{pmatrix} a \\ b \end{pmatrix}, \qquad b_{obs} = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 4 \end{pmatrix},$$

which has no solution, since no line passes through all four points. Form the normal equations:

$$A^tA = \begin{pmatrix} 4 & 10 \\ 10 & 30 \end{pmatrix}, \qquad A^tb_{obs} = \begin{pmatrix} 12 \\ 35 \end{pmatrix},$$

so $4a + 10b = 12$ and $10a + 30b = 35$. Eliminating gives $5b = 5$, hence $b = 1$ and $a = \frac{1}{2}$. The fitted line is

$$y = \tfrac{1}{2} + x.$$

Its predictions at $x = 1, 2, 3, 4$ are $1.5, 2.5, 3.5, 4.5$, so the residuals are $-0.5, +0.5, +0.5, -0.5$. They sum to zero, which is the first normal equation, and their inner product with $(1, 2, 3, 4)$ is $-0.5 + 1 + 1.5 - 2 = 0$, which is the second. The sum of their squares is 1, and by Theorem 13.4 no other line does better.

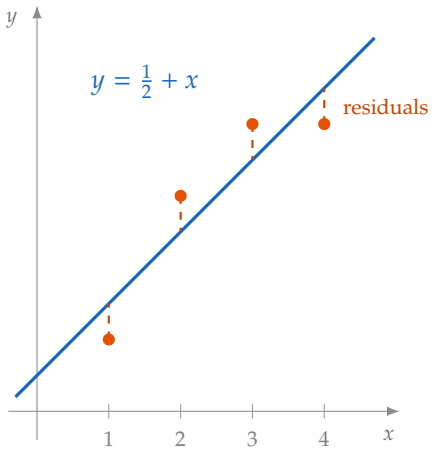


Figure 13.2. Example 13.10. No line passes through all four points, so we take the one whose vertical misses have the smallest total square. The four dashed gaps here are $-\frac{1}{2}, +\frac{1}{2}, +\frac{1}{2}, -\frac{1}{2}$, and the fact that they sum to zero and are also uncorrelated with x is precisely the pair of normal equations.

Insight

This is linear regression, and the vocabulary translates directly: the columns of A are the predictors, $\hat{x}$ is the vector of fitted coefficients, and $A\hat{x}$ is the vector of predictions. When the predictors are nearly dependent, A^tA is nearly singular and the coefficients become wildly sensitive to the data, which is the phenomenon statisticians call collinearity and Chapter 5 called an unidentifiable direction. Ridge regression replaces A^tA by $A^tA + \lambda\mathbf{1}$, which Lemma 13.5 shows is invertible for every $\lambda > 0$ whatever the data does, and which biases the answer towards small coefficients. The statistical story and the linear algebra story are the same story.

13.4 Computing it

Caution

Do not solve least squares by forming A^tA. Squaring a matrix squares its condition number, so the normal equations can be far worse conditioned than the original problem, and information present in A can be destroyed before the solve begins.

The standard illustration is

$$A = \begin{pmatrix} 1 & 1 \\ \varepsilon & 0 \\ 0 & \varepsilon \end{pmatrix}, \qquad A^t A = \begin{pmatrix} 1+\varepsilon^2 & 1 \\ 1 & 1+\varepsilon^2 \end{pmatrix}.$$

The columns of A are independent for every $\varepsilon \neq 0$. But if ε is small enough that $1+\varepsilon^2$ rounds to 1, and on a machine with double precision that means ε below about 10^{-8}, then the computed A^tA is exactly $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, which is singular. The problem was well posed and the first step of the algorithm destroyed it.

Proposition 13.11 (Least squares by QR). *Let A have independent columns and let $A = QR$ be its factorisation from Theorem 12.12. Then the least squares solution is obtained by solving the triangular system*

$$R\,\hat{x} = Q^t b.$$

Proof. Substituting $A = QR$ into the normal equations and using $Q^tQ = \mathbf{1}$,

$$A^tA\hat{x} = R^tQ^tQR\,\hat{x} = R^tR\,\hat{x}, \qquad A^tb = R^tQ^tb.$$

So the normal equations read $R^tR\hat{x} = R^tQ^tb$. Since R is upper triangular with positive diagonal it is invertible, hence so is R^t, and cancelling it gives the claim. □

Key Idea

Notice what the proof did: it produced the same answer without ever forming A^tA. The matrix R^t appears on both sides and cancels symbolically, so the numerically dangerous product never has to exist. Then $R\hat{x} = Q^tb$ is a triangular system, solved by back substitution at $O(n^2)$. This is what a library does when you ask it to fit a model, and the whole cost is the QR factorisation.

13.5 The four fundamental subspaces

We can now assemble everything Chapters 5 to 6 produced into one picture.

Theorem 13.12 (The big picture). *Let A be $m \times n$ of rank r. Then*

$$\mathbb{R}^n = R(A) \oplus N(A), \qquad \mathbb{R}^m = C(A) \oplus N(A^t),$$

where $\oplus$ records, as in Theorem 13.2, that every vector splits into one part from each summand in exactly one way. In each case the two subspaces are orthogonal complements, and the dimensions are

$$\dim R(A) = \dim C(A) = r, \qquad \dim N(A) = n - r, \qquad \dim N(A^t) = m - r.$$

Moreover A maps $R(A)$ one-to-one and onto $C(A)$.

Proof. The orthogonality and the splittings are Lemma 13.8 with Theorem 13.2. The dimensions of $R(A)$, $C(A)$ and $N(A)$ are Theorem 6.15 and Corollary 6.17; applying the same to A^t, whose rank is also r, gives $\dim N(A^t) = m - r$.

For the last statement, restrict A to $R(A)$. Its kernel is $R(A) \cap N(A) = R(A) \cap R(A)^\perp = \{\mathbf{0}\}$, so the restriction is injective; its image lies in $C(A)$ and has dimension $r = \dim C(A)$ by rank and nullity, so it is all of $C(A)$. □

Figure 13.3 arranges the four spaces as the theorem describes them: the source splits at right angles into the row space and the null space, the target into the column space and the left null space, and between the two upper halves A is a bijection.

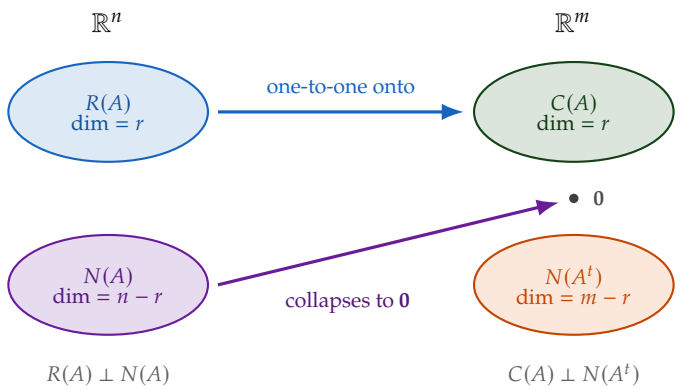


Figure 13.3. Theorem 13.12. The source splits into the row space and the null space, at right angles; the target splits into the column space and the left null space, likewise. Between the two upper halves A is a bijection, and the lower left half is crushed to the origin. Solving $Ax = b$ asks whether b lies in the upper right blob; least squares replaces b by its projection there when it does not.

Key Idea

Read least squares off the picture. The vector b sits somewhere in $\mathbb{R}^m$ and splits as a part in $C(A)$ and a part in $N(A^t)$. The first part is reachable and the second is not, so the best possible is to hit the first exactly, which is what Theorem 13.9 does; the residual that remains is the $N(A^t)$ part, and it is orthogonal to everything A can produce. The system is consistent precisely when that part is zero.

Open Problem 13.13 (Sketching without randomness)**.** *A least squares problem with n equations and $d \ll n$ unknowns can be solved to relative accuracy ε in time close to the number of nonzeros of A, by first multiplying by a random $O(d/\varepsilon^2) \times n$ matrix S and solving the small problem instead.* Is there a deterministic S, computable in nearly the same time, with the same guarantee?

What is known. *Random constructions are abundant and near optimal: a subsampled Hadamard transform, or a sparse embedding with one nonzero per column, gives a subspace embedding with $O(d/\varepsilon^2)$ rows and can be applied in time close to $O(nnz(A))$ [654]. Deterministic constructions exist but are either far larger or far slower: the best known deterministic subspace embeddings either take time polynomial in n and d to build, or use substantially more than d/ε^2 rows. Closing that gap is open.*

Where the difficulty sits. *A subspace embedding must preserve the norm of* every *vector in a d-dimensional subspace, and the subspace is unknown until A arrives. A random matrix handles all subspaces at once because a union bound over a net costs only a logarithm; a deterministic matrix must be verified against all of them, and there is no known certificate short enough. This is the same wall as the explicit restricted isometry problem of Chapter 55, and the two are closely related.*

Where a reader might start. *Take $d = 1$, so the subspace is a line, and construct a deterministic S with few rows preserving the norm of every vector on it. That is easy. Now take $d = 2$ and require the guarantee for every line in a plane you do not know in advance, and see the problem change character. Chapter 24 builds the random construction, and Chapter 42 shows what leverage scores buy when the matrix is available in advance.*

Example 13.14 (A line through three points, in fractions)**.** *Fit $y = c_0 + c_1 t$ to the data $(0,1)$, $(1,3)$, $(2,2)$. The design matrix and normal equations of Lemma 13.5 are*

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad X^t X = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \qquad X^t y = \begin{pmatrix} 6 \\ 7 \end{pmatrix},$$

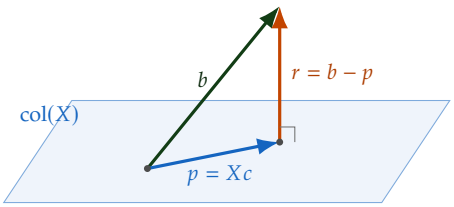
and solving gives $c_0 = \frac{3}{2}$, $c_1 = \frac{1}{2}$ exactly. The fitted values are $\frac{3}{2}, 2, \frac{5}{2}$ and the residuals

are $-\frac{1}{2}, 1, -\frac{1}{2}$.
Now check the geometry rather than the algebra. The residual must be orthogonal to every *column of* X*, which is Theorem 13.2, and it is:*

$$\langle r, \mathbf{1} \rangle = -\tfrac{1}{2} + 1 - \tfrac{1}{2} = 0, \qquad \langle r, t \rangle = 0 \cdot (-\tfrac{1}{2}) + 1 \cdot 1 + 2 \cdot (-\tfrac{1}{2}) = 0.$$

Two equations, two unknowns, and they are exactly the two orthogonality conditions. The normal equations are not a computational trick; they are the statement that the error points out of the plane you were allowed to search. *The residual sum of squares is* $\frac{1}{4} + 1 + \frac{1}{4} = \frac{3}{2}$*, and no line does better, by Theorem 13.4.*

Figure 13.4 draws that sentence: the data vector does not lie in the column space, the best available answer is the nearest point of it, and the residual leaves at a right angle.



$\langle r, a_1 \rangle = \langle r, a_2 \rangle = 0$, and $\|p\|^2 + \|r\|^2 = 12.5 + 1.5 = 14 = \|b\|^2$

Figure 13.4. Least squares is a right angle. The data vector b does not lie in the column space of X, so no exact solution exists; the best available answer is the point p of that subspace nearest to b, and Theorem 13.2 says the error $r = b - p$ is perpendicular to the whole subspace. That perpendicularity is the normal equations: one orthogonality condition per column, which is exactly as many equations as there are unknowns. Pythagoras then splits the data into the part that was explainable and the part that was not, and Example 13.14 carries out the arithmetic, with $b = (1, 3, 2)$, $p = (\frac{3}{2}, 2, \frac{5}{2})$ and $r = (-\frac{1}{2}, 1, -\frac{1}{2})$.

Exercises

Exercise 13.15. *In* $\mathbb{R}^4$*, let* $W = \operatorname{span}\{(1, 1, 0, 0), (0, 1, 1, 0)\}$*. Find a basis of* $W^\perp$ *and check that the dimensions add to* 4*.*

Exercise 13.16. *Let $W = \operatorname{span}\{(1,1,0),(0,1,1)\} \subseteq \mathbb{R}^3$ and $v = (1,0,0)$. Compute $P_W v$ two ways: by orthonormalising a basis of W and using Definition 13.3, and by the formula $A(A^tA)^{-1}A^t v$. Check that $v - P_W v$ is orthogonal to both spanning vectors.*

Exercise 13.17. *Fit a line by least squares to the points $(1,2)$, $(2,2)$, $(3,4)$, $(4,5)$, and compute the residuals. Verify that they sum to zero and that their inner product with $(1,2,3,4)$ is zero.*

Exercise 13.18. *Fit $y = a + bx + cx^2$ to the three points $(-1,1)$, $(0,0)$, $(1,1)$ by least squares. Why is the answer exact here, and what does that say about $C(A)$?*

Exercise 13.19. *Show that if $P^t = P$ and $P^2 = P$ then P is the orthogonal projection onto $C(P)$. Hint: show $v - Pv$ is orthogonal to every Pw. Then give a 2×2 matrix with $P^2 = P$ that is not an orthogonal projection, and describe what it does.*

Exercise 13.20. *Take $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $b = (1,2,3)$. Solve the least squares problem twice, once through the normal equations and once through $R\hat{x} = Q^t b$ using the factorisation from Exercise 12.21, and confirm the answers agree.*

Exercise 13.21. *Find the distance from the point $(1,2,3)$ to the plane $x + y + z = 0$ in $\mathbb{R}^3$, using Theorem 13.4 rather than a formula from coordinate geometry.*

Exercise 13.22. *Let $A = \begin{pmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{pmatrix}$ and $b = (1,2,4)$. Show that the normal equations are consistent but have infinitely many solutions, find them all, and identify which one has the smallest norm.* This one is the pseudoinverse solution.

Exercise 13.23. *For $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$, find bases for all four fundamental subspaces, verify the two dimension counts of Theorem 13.12, and check that each pair is orthogonal.*

References and further reading

The Gauss and Legendre dispute is well documented and worth reading about; Stigler's history of statistics is the standard account and takes Legendre's side on publication and Gauss's on priority. The treatment here, deriving least squares from the best-approximation theorem rather than by differentiating a sum of squares, is the one that generalises, and it is the reason the normal equations come out as a statement about orthogonality rather than as a stationarity condition. Austin [35] is particularly good on the data-fitting side and on the picture in Figure 13.2; Hefferon [307] gives more worked projections. Axler [37] proves the best-approximation theorem in the same way we have and then uses it for polynomial approximation of functions, which is worth seeing because it shows the theorem has nothing to do with data. Banica [51], our baseline, treats projections as the self-adjoint idempotents of Exercise 13.19, which is the operator-algebra point of view and the right one for what comes later. On the numerical side Trefethen and Bau [607] devote a lecture apiece to the conditioning of least squares and to why the normal equations are the wrong algorithm; the example in the Caution of Section 13.4 is theirs.